WORKED SOLUTION

STEP II 2010, Question 12

Probability and Statistics • Mean and Median

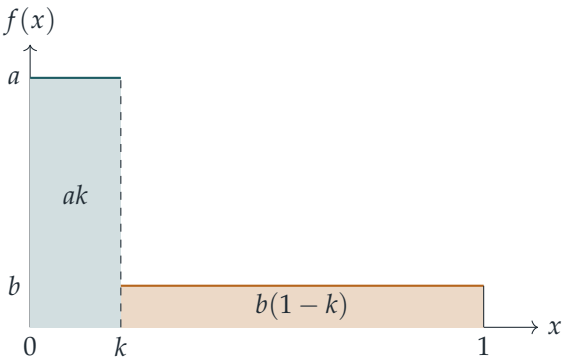
STEP II 2010, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q12

The key idea

The density is a two-step function, so every probability is an area of a rectangle and the only real condition is that the total area is 1. That condition fixes k in terms of a and b, and once k is eliminated everything is algebra. For the median, the question is simply whether the tall first rectangle already holds half of the probability.



Solution

The two rectangles in the sketch have areas ak and $b(1-k)$, so the density integrates to 1 exactly when

$$ak + b(1-k) = 1, \qquad \text{that is,} \qquad k = \frac{1-b}{a-b}.$$

(We may divide by $a-b$ because $a > b$.) We use this value of k throughout.

Why $a > 1$ and $b < 1$. The equation $ak + b(1-k) = 1$ says that 1 is a weighted average of a and b, with weights k and $1-k$, both strictly between 0 and 1. A weighted average with positive weights lies strictly between the two values, so $b < 1 < a$. To put it without words: $1 - b = k(a-b) > 0$ and $a - 1 = (1-k)(a-b) > 0$.

Part (i)

Integrating $xf(x)$ over each piece,

$$\mathrm{E}(X) = \int_0^k ax\,\mathrm{d}x + \int_k^1 bx\,\mathrm{d}x = \tfrac{1}{2}ak^2 + \tfrac{1}{2}b(1-k^2) = \tfrac{1}{2}(b + (a-b)k^2).$$

Now $(a-b)k^2 = \dfrac{(1-b)^2}{a-b}$, so

$$\mathrm{E}(X) = \frac{b(a-b)+(1-b)^2}{2(a-b)} = \frac{ab - b^2 + 1 - 2b + b^2}{2(a-b)} = \frac{1-2b+ab}{2(a-b)},$$

as required.

Part (ii)

The cumulative distribution function is

$$\mathrm{F}(x) = \begin{cases} ax & 0 \le x < k, \\ ak + b(x-k) & k \le x \le 1. \end{cases}$$

It is continuous and strictly increasing on $[0,1]$, so the median M is the unique solution of $\mathrm{F}(M) = \frac{1}{2}$. Which formula applies depends on whether the first rectangle holds at least half of the probability.

> **Ask yourself** How do we turn "the first rectangle has area at least $\frac{1}{2}$" into a condition on a and b?

Substitute for k and clear the positive denominator $a-b$:

$$ak \ge \tfrac{1}{2} \iff 2a(1-b) \ge a-b \iff 2a - 2ab \ge a - b \iff a + b \ge 2ab.$$

Case $a+b \ge 2ab$. Then $\mathrm{F}(k) = ak \ge \frac{1}{2}$, so the median lies in $[0,k]$ and solves $aM = \frac{1}{2}$:

$$M = \frac{1}{2a}.$$

Case $a+b \le 2ab$. Then $ak \le \frac{1}{2}$, so the median lies in $[k,1]$. It is easiest to work from the right-hand end: the probability to the right of M is $b(1-M)$, and this must equal $\frac{1}{2}$. Hence

$$M = 1 - \frac{1}{2b} = \frac{2b-1}{2b}.$$

When $a+b = 2ab$ both formulae apply and agree, since then $1 - \frac{1}{2b} = \frac{2ab-a}{2ab} = \frac{b}{2ab} = \frac{1}{2a}$.

Part (iii)

We compute $\mathrm{E}(X) - M$ in each case and show it is positive.

Case $a+b \geq 2ab$. Over the common denominator $2a(a-b)$,

$$\mathrm{E}(X) - \frac{1}{2a} = \frac{a(1-2b+ab)-(a-b)}{2a(a-b)} = \frac{b-2ab+a^2b}{2a(a-b)} = \frac{b(a-1)^2}{2a(a-b)}.$$

Case $a+b \leq 2ab$. Over the common denominator $2b(a-b)$,

$$\begin{aligned}\mathrm{E}(X) - \frac{2b-1}{2b} &= \frac{b(1-2b+ab)-(2b-1)(a-b)}{2b(a-b)} \\ &= \frac{(b-2b^2+ab^2)-(2ab-2b^2-a+b)}{2b(a-b)} = \frac{a(1-b)^2}{2b(a-b)}.\end{aligned}$$

Ask yourself Why are these strictly positive, and not merely non-negative?

In both cases a, b and $a-b$ are positive, and the squares are strictly positive because we showed at the start that $a > 1$ and $b < 1$. So $\mathrm{E}(X) - M > 0$ in every case, and $M < \mathrm{E}(X)$.

Checks

- Take $a = 3$, $b = \frac{1}{2}$: then $k = \frac{1}{5}$, $a+b = \frac{7}{2} \geq 3 = 2ab$, and $\mathrm{E}(X) = \frac{1-1+\frac{3}{2}}{5} = 0.3$ while $M = \frac{1}{6}$. The difference $0.1\overline{3}$ matches $\frac{b(a-1)^2}{2a(a-b)} = \frac{2}{15}$.
- Take $a = \frac{5}{4}$, $b = \frac{9}{10}$: now $a+b = 2.15 \leq 2.25 = 2ab$, so $M = 1 - \frac{5}{9} = \frac{4}{9} \approx 0.444$, and $\mathrm{E}(X) = \frac{13}{28} \approx 0.464$.
- As $a \to 1^+$ (so $b \to 1^-$) the density tends to the uniform one, and both differences tend to 0: mean and median coincide at $\frac{1}{2}$, as they should by symmetry.

We checked numerically, for 300 random pairs (a, b), that the median found by bisection on F matches the case formula, that the mean by midpoint quadrature matches part (i), and that $\mathrm{E}(X) - M$ equals the factorised expressions above.

Common slips

- Forgetting to eliminate k: the answers are wanted in terms of a and b only.
- Writing the second median as the solution of $b(M-k) = \frac{1}{2}$, forgetting the probability ak already accumulated on $[0, k)$.
- Concluding only $\mathrm{E}(X) \geq M$: the strict inequality needs $a \neq 1$ and $b \neq 1$, which is exactly what the opening part provides.

Beyond STEP

This density is decreasing, so it is "skewed to the right": its long tail pulls the mean beyond the median. That pattern is common but not universal, and a first course in probability shows how mean, median and mode relate for unimodal distributions and where the familiar rules of thumb fail. Step functions like this one are also the building blocks of histograms, and of the simple functions used to define the integral in measure theory. For more on continuous distributions, see Grimmett and Welsh, *Probability: An Introduction*.

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